CURRICULUM VITAE

Qian Ouyang

Machine learning for materials and molecular simulation.

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Education

Xi'an Jiaotong University

Fourth-year undergraduate | Artificial Intelligence

Artificial Intelligence Experimental Class

2023 - 2027 (expected)

Academic experience

Research Assistant October 2025 - Present

University of Notre Dame

Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

Summer Research Participant (iSURE) July - September 2026

University of Notre Dame

International Summer Undergraduate Research Experience (iSURE).

Visiting Student August - December 2025

University of Notre Dame

Research interests

AI for Materials; Molecular Dynamics & Machine Learning Potentials; HPC Optimization of ML Interatomic Potentials; Coarse-Grained Molecular Modeling; Representation Learning / JEPA; Polymer Property Prediction.

Selected research

Molecular representations and physical prediction

From CG polymer and MACE water explorations to controlled physical-response tests and current pretrained-feature transfer studies; each task has its own physical targets and evidence limits.

Pretrained atomistic representations: transfer and label efficiency

Ongoing research · crystal study in preparation

Compared pretrained and scratch Atom-JEPA models on QM9 under controlled data and update budgets. Current work prepares crystal-property and label-efficiency studies; crystal training and transfer results remain unestablished.

State-space models for controlled physical response

Exploratory study completed | Closed 24 October 2026

Compared state-space and recurrent models on synthetic hysteretic dynamics, separating trajectory fit from control-response prediction. Curve-level gains did not resolve the decisive response contrast; real-material validation remains open.

Selected research (continued)

MLIP material properties and efficiency

Ongoing polymer-density workflow development, supported by earlier MACE resource measurements and the UMA / FAIR-Chem exploration that motivated my interest in MLIP efficiency.

Polymer density workflows with MACE

Ongoing workflow development | Displayed pilot: 17 September 2026

Developing polymer-density workflows with classical preparation and MACE molecular dynamics, GAFF2/reference comparisons, chemical-identity checks, sampling QC, and resource-aware HPC execution. Displayed September results remain preliminary.

GPU resource trade-offs in MACE molecular dynamics

Engineering pilot | September 2026

Evaluated runtime, sampled per-GPU memory, and GPU-hour trade-offs for a fixed MACE polymer simulation workload. Engineering pilot with one short run per configuration; not a general scaling claim.

Polymer elastic-response workflows with UMA / FAIR-Chem

Resource-limited exploration

Explored UMA / FAIR-Chem polymer calculations with LAMMPS. GPU-memory constraints limited progress before validated property results and motivated my interest in MLIP efficiency.

Technical skills

Molecular simulation & MLIPs: Experience with UMA/FAIR-Chem and MACE for molecular simulation.

High-performance computing: Experience using Notre Dame's Center for Research Computing (CRC) for research computing.

Programming languages: C++, Python